

On the Correctness of Real Parameterization, the Modular Caveat, and Conditional Deduction under the Global Binary Normalization Hypothesis

Statement of the Problem

Consider the equalities

$$\begin{aligned}x^n &= (m^n - p^n)^n, \\z^n &= (m^n + p^n)^n, \\y^n &= 2n l^n,\end{aligned}$$

where the parameters

$$m, p, l$$

are not required to be integers, but can be real numbers.

Furthermore, a subsequent conditional step is considered: if we accept as a separate structural hypothesis the equality

$$o(n) = 2,$$

is the following deduction correct

$$2^n = 2n$$

and, consequently,

$$n = 1 \quad \text{or} \quad n = 2.$$

In version v13 of the paper, this hypothesis is specified as the *global binary normalization hypothesis*. Its meaning is not that the local parameter $q(j)$ equals 2 for all $j > 1$, but rather that the symmetric boundary value

$$\lim_{j \rightarrow 1+} q(j) = 2$$

is used as the normalizing value for the global scale parameter

$$o(n) = 2.$$

In other words, two distinct objects are differentiated:

$$q(j)$$

as a local binomial scale depending on the ratio

$$j = \frac{m}{p},$$

and

$$o(n)$$

as a globally normalized scale depending on the exponent n .

The new version of the formalization additionally takes into account the modular caveat. It is necessary in order to explicitly separate equalities over

$$\mathbb{N}, \quad \mathbb{Z}, \quad \mathbb{Q}, \quad \mathbb{R}$$

from modular congruences and equations over finite fields

$$\mathbb{F}_q.$$

In other words, four distinct levels are considered:

the real level,

the integer level,

the modular level,

the global normalization level.

Real parameterization is used to reconstruct the algebraic scheme. The integer level is needed for the connection with Fermat's Last Theorem. The modular level is not intentionally used as a proof method, but it is separately explained why modular solutions of congruences do not constitute counterexamples to the integer formulation. The global normalization level fixes the additional structural hypothesis

$$o(n) = 2,$$

without which the final prohibition of exponents $n > 2$ is not an unconditional deduction from the real parameterization alone.

1. Correctness of the Equalities for x^n and z^n

The equalities

$$x^n = (m^n - p^n)^n$$

and

$$z^n = (m^n + p^n)^n$$

are correct provided that it is preliminarily set

$$x = m^n - p^n,$$

$$z = m^n + p^n.$$

This is the standard raising of equal quantities to the same power.

In other words, if

$$x = m^n - p^n,$$

then automatically

$$x^n = (m^n - p^n)^n.$$

Similarly, if

$$z = m^n + p^n,$$

then automatically

$$z^n = (m^n + p^n)^n.$$

Here it is not required for m and p to be integers. It is sufficient for the corresponding expressions to be defined within the considered numerical domain, such as the domain of real numbers.

However, it is important to clarify the following. If x, z are assumed in advance to be natural numbers, then the equalities

$$x = m^n - p^n, \quad z = m^n + p^n$$

become consistency conditions between the real parameterization and the natural values of x, z . In the general real case, there are no such arithmetic restrictions; in the integer case, additional conditions of parity and perfect powers emerge.

In Coq v13, this part corresponds to several levels of formalization.

First, the lemma

$$\text{sum_diff_from_parameters_R}$$

fixes the real identities:

$$z = m^n + p^n, \quad x = m^n - p^n$$

imply

$$z + x = 2m^n, \quad z - x = 2p^n.$$

Second, the lemma

$$\text{parameter_equations_raise_nat}$$

verifies an elementary fact: if at the natural number level we are given

$$z = m^n + p^n, \quad x = m^n - p^n,$$

then after raising to the power, we automatically obtain

$$z^n = (m^n + p^n)^n, \quad x^n = (m^n - p^n)^n.$$

Thus, at this step, there is no hidden mathematical assumption: only the elementary property of equality is used.

2. Correctness of the Equality $y^n = 2n l^n$ for Real l

The equality

$$y^n = 2n l^n$$

is correct if l is understood as a real scaling parameter.

After the binomial expansion of the expression

$$(m^n + p^n)^n - (m^n - p^n)^n$$

a sum exclusively over the odd binomial terms is obtained:

$$(m^n + p^n)^n - (m^n - p^n)^n = 2 \sum_{\substack{1 \leq j \leq n \\ j \text{ odd}}} \binom{n}{j} (m^n)^{n-j} (p^n)^j.$$

Let us denote the inner sum by

$$S = \sum_{\substack{1 \leq j \leq n \\ j \text{ odd}}} \binom{n}{j} (m^n)^{n-j} (p^n)^j.$$

Then

$$y^n = 2S.$$

If we now define the real parameter l by the condition

$$l^n = \frac{S}{n},$$

we obtain

$$S = n l^n.$$

Consequently,

$$y^n = 2S = 2n l^n.$$

Thus, the equality

$$y^n = 2n l^n$$

is correct as a real representation when defining l through

$$l^n = \frac{1}{n} \sum_{\substack{1 \leq j \leq n \\ j \text{ odd}}} \binom{n}{j} (m^n)^{n-j} (p^n)^j.$$

In this context, l is not required to be an integer.

If n is odd, the real root of S/n exists for any real value of S/n . If n is even, then for a real positive l , one must choose a situation where

$$\frac{S}{n} > 0.$$

It is precisely this positive case that is formalized in Coq v13 through the positivity of the parameter l .

In Coq v13, this specification is reflected by the definition

$$\text{real_core_factorization } n \text{ core } l,$$

which means that for $n > 0$

$$\text{core} = n l^n$$

in the real domain. This is not a statement about integer divisibility, but specifically a real factorization.

Furthermore, the lemma

$$\text{real_core_factorization_substitutes_y_power}$$

shows that if

$$y^n = 2 \text{ core}$$

and

$$\text{core} = n l^n,$$

then

$$y^n = 2n l^n.$$

This corresponds exactly to the transition from the binomial core to the scaling equality in the paper.

3. The Important Distinction from Integer Divisibility

If l is treated as a real number, there is no need to assert that the sum

$$S = \sum_{\substack{1 \leq j \leq n \\ j \text{ odd}}} \binom{n}{j} (m^n)^{n-j} (p^n)^j$$

is divisible by n in integers.

That is a different, stronger statement.

For real parameterization, it is sufficient to define

$$l^n = \frac{S}{n}.$$

Therefore, a potential problem with integer divisibility does not collapse the real scheme, provided it is explicitly stated that

$$l \in \mathbb{R}.$$

Moreover, it is important to understand that the assertion that the odd binomial core is always divisible by n in integers is generally false. For example, for

$$n = 6, \quad m = 1, \quad p = 1$$

the odd core takes the form

$$\binom{6}{1} + \binom{6}{3} + \binom{6}{5} = 6 + 20 + 6 = 32,$$

and the number 32 is not divisible by 6 in integers.

This does not collapse the real scheme, because it does not claim the integer divisibility of the core by n . The real scheme uses the definition

$$l^n = \frac{S}{n},$$

rather than the statement

$$n \mid S$$

in the ring of integers.

Consequently, one must strictly distinguish between:

$$l^n = \frac{S}{n}$$

as a real definition, and

$$n \mid S$$

as an integer arithmetic statement.

The former is permissible in the real model. The latter requires a separate proof and is generally incorrect.

In paper v13, this is already expressed by stating that the expression allows factoring out the common multiplier n in real algebraic factorization. In other words, it is not about divisibility in $\mathbb{Z}$, but about the representation

$$S = n l^n$$

in the real domain.

Coq v13 is aligned precisely with this cautious wording: the definition

real_core_factorization

does not state that n is an integer divisor of the core; it states that the core is represented in the form

$$\text{core} = n l^n$$

over $\mathbb{R}$.

4. Transition to the Scaling Parameter o

If we introduce the scaling parameter o through the equality

$$y = o l,$$

then for $l \neq 0$ we have

$$y^n = o^n l^n.$$

On the other hand, from the previous equality

$$y^n = 2n l^n.$$

Consequently,

$$o^n l^n = 2n l^n.$$

If

$$l \neq 0,$$

we can cancel l^n to obtain

$$o^n = 2n.$$

In other words,

$$o = (2n)^{1/n}$$

choosing the positive real branch of the root.

In Coq, this corresponds to the definition

covers_with o n ,

which means

$$o^n = 2n.$$

More precisely, the Coq lemma

positive_l_scaling_forces_cover

asserts the following. If

$$n > 0,$$

$$l > 0,$$

$$y = o l,$$

and

$$y^n = 2n l^n,$$

then necessarily

$$o^n = 2n.$$

This is exactly the transition used in the paper: from the real scaling equality to the cover condition

$$\text{covers_with } o\ n.$$

Also, in Coq v13, the structure

PositiveRealManuscriptData

is introduced, where the parameters

$$m, p, l, x, y, z$$

are considered real numbers, with m, p, l being positive, and the equalities

$$x^n = (m^n - p^n)^n,$$

$$z^n = (m^n + p^n)^n,$$

$$y^n = 2n\ l^n$$

are provided as fields of this structure.

The lemma

positive_real_record_scaling_forces_cover

then shows that if additionally

$$y = o\ l,$$

then the structure data imply

$$\text{covers_with } o\ n.$$

5. Local Scale $q(j)$ and Boundary Normalization

In version v13 of the paper, the relationship between the local binomial scale and the global parameter $o(n)$ is significantly clarified.

Let

$$m = j\ p, \quad j > 1.$$

Then

$$m^n = j^n p^n.$$

Substitution into the difference

$$(m^n + p^n)^n - (m^n - p^n)^n$$

yields

$$(p^n)^n ((j^n + 1)^n - (j^n - 1)^n).$$

By the binomial formula

$$(j^n + 1)^n - (j^n - 1)^n = 2 \sum_{\substack{1 \leq k \leq n \\ k \text{ odd}}} \binom{n}{k} (j^n)^{n-k}.$$

Let us define the local parameter $q(j)$ by the condition

$$q(j)^n := 2 \sum_{\substack{1 \leq k \leq n \\ k \text{ odd}}} \binom{n}{k} (j^n)^{n-k}.$$

Then we obtain the relation

$$\left(\frac{p^n q(j)}{l}\right)^n = 2n.$$

From this, the locally represented scale is naturally introduced

$$o(n) = \frac{p^n q(j)}{l}.$$

However, it is important to emphasize:

$$q(j) \neq o(n).$$

The function $q(j)$ is a local binomial scale depending on the ratio

$$j = \frac{m}{p}.$$

In the working non-degenerate region

$$j > 1$$

the point

$$j = 1$$

does not belong to the domain of non-trivial solutions, because for $j = 1$

$$m = p$$

and therefore

$$x = m^n - p^n = 0.$$

Consequently, $j = 1$ is a boundary degenerate point.

Nevertheless, one can consider the continuous boundary value of the local scale. As

$$j \rightarrow 1+$$

we have

$$\lim_{j \rightarrow 1+} q(j)^n = 2 \sum_{\substack{1 \leq k \leq n \\ k \text{ odd}}} \binom{n}{k}.$$

From the standard binomial identity

$$\sum_{\substack{1 \leq k \leq n \\ k \text{ odd}}} \binom{n}{k} = 2^{n-1}$$

we obtain

$$\lim_{j \rightarrow 1+} q(j)^n = 2 \cdot 2^{n-1} = 2^n.$$

Consequently,

$$\lim_{j \rightarrow 1+} q(j) = 2$$

for the positive real branch of the root.

Inside the region $j > 1$, the local parameter $q(j)$ generally depends on j and, assuming positive terms, is strictly greater than the boundary value. Therefore, paper v13 does not assert that

$$q(j) = 2$$

for all

$$j > 1.$$

It asserts something else: the symmetric boundary value

$$\lim_{j \rightarrow 1+} q(j) = 2$$

is used as the normalizing value for the global binary scale.

In other words,

$$o(n) := \lim_{j \rightarrow 1+} q(j) = 2$$

is not a pointwise identification of the local function $q(j)$ with $o(n)$, but a step of global normalization.

In Coq v13, this is reflected by the structure

LocalScaleGlobalNormalization.

It contains:

$$n,$$

$$j,$$

$$q,$$

$$o,$$

the non-degeneracy condition

$$1 < j,$$

the boundary equation

$$\text{boundary_q_equation } n \ 2,$$

and a separate global normalization field

$$o = 2.$$

The lemma

local_scale_record_uses_global_boundary

shows that such a structure indeed yields two key facts:

$$o = 2$$

and

$$\text{boundary_q_equation } n \ 2.$$

Thus, Coq v13 does not analytically prove the entire limit

$$\lim_{j \rightarrow 1+} q(j) = 2$$

from the binomial sum, but carefully establishes it as a boundary-normalizing structural component of the model.

6. The Conditional Step $o(n) = 2$

Next, a separate global binary normalization hypothesis is introduced:

$$o(n) = 2.$$

It is crucial to emphasize that this is specifically a structural hypothesis, not an automatically proven fact derived solely from real parameterization.

Accepting this hypothesis, from

$$o(n)^n = 2n$$

we obtain

$$2^n = 2n.$$

This is a purely elementary exponential-linear equation.

It is equivalent to

$$n = 2^{n-1}.$$

Its positive integer solutions are:

$$n = 1$$

and

$$n = 2.$$

For all

$$n > 2$$

we have a strict inequality

$$2^n > 2n.$$

Consequently, under the condition

$$o(n) = 2$$

the exponents

$$n > 2$$

are ruled out.

This conditional step is the central point of the model. Coq v13 rigorously verifies that from

$$o(n) = 2$$

and

$$o(n)^n = 2n$$

it follows that

$$n \in \{1, 2\}.$$

In Coq v13, this is expressed through the lemmas:

`covers_two_only_small,`

`covers_with_two_characterisation,`

`binary_scaling_roots_only_one_two.`

Moreover, Coq v13 verifies the equivalent form

$$2^n = 2n \iff n = 2^{n-1}$$

in the lemma

$$\text{two_pow_eq_two_mul_iff_shift.}$$

However, Coq v13 does not unconditionally prove that

$$o(n) = 2$$

follows from real parameterization alone. This condition remains the structural hypothesis of global binary normalization.

7. The Modular Caveat

Paper version v13 and Coq v13 separately address the modular objection.

The present model does not construct a proof via modular arithmetic and does not assert the absence of solutions for the corresponding congruences in finite fields. The considered scheme pertains to equalities over

$$\mathbb{N}, \quad \mathbb{Z}, \quad \mathbb{Q}, \quad \mathbb{R},$$

not to equations over a finite field

$$\mathbb{F}_q.$$

In particular, the integer equality

$$x^n + y^n = z^n$$

indeed implies a congruence modulo any q :

$$x^n + y^n \equiv z^n \pmod{q}.$$

However, the converse is generally false: the congruence

$$x^n + y^n \equiv z^n \pmod{q}$$

does not imply the equality

$$x^n + y^n = z^n$$

in integers.

In other words, a modular congruence merely means that there exists some integer t for which

$$z^n = x^n + y^n + qt.$$

An integer solution to Fermat's equation corresponds to the special case

$$t = 0.$$

Therefore, the existence of solutions to the congruences

$$x^n + y^n \equiv z^n \pmod{q}$$

for

$$n > 2$$

does not constitute a counterexample to Fermat's Last Theorem and does not refute the real parameterization used in this model.

If one specifically transitions to modulo q , the parameters must be defined separately at the modular level. For example, instead of the real equalities

$$z = m^n + p^n, \quad x = m^n - p^n$$

one must separately fix the residues

$$m^n \equiv A \pmod{q}, \quad p^n \equiv B \pmod{q}.$$

Subsequently, the corresponding modular relations take the form

$$z \equiv A + B \pmod{q}, \quad x \equiv A - B \pmod{q}.$$

Thus, real parameterization and modular arithmetic belong to different levels of description. This paper uses the first level: equalities over

$$\mathbb{N}, \quad \mathbb{Z}, \quad \mathbb{Q}, \quad \mathbb{R},$$

rather than a distinct theory of solutions over finite fields.

8. How the Modular Caveat is Reflected in Coq v13

Coq v13 features a dedicated section

Section ModularRemark.

In it, the relation of modular congruence is defined as follows:

$$a \equiv b \pmod{q}$$

means that there exists an integer t such that

$$a = b + qt.$$

In Coq language, this is defined as

modular_congruent q a b .

Next, the key propositions are formalized.

First, if the equality of powers

$$x^n + y^n = z^n,$$

holds in integers, then the corresponding congruence modulo any q is automatically satisfied:

$$z^n \equiv x^n + y^n \pmod{q}.$$

This corresponds to the lemma

integer_equality_implies_modular_power_congruence.

The logic of the proof is simple: given a true equality, one can take

$$t = 0.$$

Second, Coq explicitly establishes that equality in integers corresponds to a zero modular witness:

$$t = 0.$$

This is reflected in the lemma

integer_equality_is_zero_modular_witness.

Third, Coq verifies a concrete example demonstrating that modular congruence is not the same as integer equality:

$$3^3 \equiv 1^3 + 1^3 \pmod{5},$$

because

$$27 = 2 + 5 \cdot 5,$$

yet

$$1^3 + 1^3 \neq 3^3$$

as an equality in integers, since

$$2 \neq 27.$$

This corresponds to the lemma

modular_congruence_not_integer_equality.

Thus, Coq v13 rigorously confirms the methodological premise of the modular caveat:

integer equality $\implies$ modular congruence,

but

modular congruence $\not\Rightarrow$ integer equality.

Furthermore, the structure

ModularParameterResidues

formalizes the idea that when transitioning to modulo q , one must separately define the residues

$$m^n \equiv A \pmod{q}, \quad p^n \equiv B \pmod{q},$$

and only then separately define

$$z \equiv A + B \pmod{q}, \quad x \equiv A - B \pmod{q}.$$

This is in complete agreement with the modular caveat: modular parameters represent a separate level of description, not a direct replacement for real parameterization.

9. The Purpose of the Verification in Coq v13

Coq v13 formalizes precisely the conditional scheme of global normalization.

It introduces a structure where the parameters

$$m, p, l, x, y, z$$

are treated as real numbers.

In this structure, the equalities

$$x^n = (m^n - p^n)^n,$$

$$z^n = (m^n + p^n)^n,$$

$$y^n = 2n l^n$$

are provided as data fields.

Then it is proved that if additionally

$$y = ol$$

and

$$l > 0,$$

it follows from

$$y^n = 2n l^n$$

that

$$o^n = 2n.$$

This corresponds exactly to the condition

$$\text{covers_with } o \ n.$$

Coq v13 also separates the real level from the integer level. For real parameters, a block is used where

$$m, p, l, x, y, z \in \mathbb{R}.$$

For the integer level, a separate structure is introduced

$$\text{PositiveIntegerManuscriptData},$$

where the parameters lie in

$$\mathbb{Z}.$$

At the integer level, additional arithmetic conditions emerge, such as parity conditions:

$$z + x \text{ is even}, \quad z - x \text{ is even}.$$

This is reflected in the lemmas

$$\text{parity_condition_Z},$$

$$\text{no_parameters_if_parity_violation},$$

$$\text{no_parameters_if_odd}.$$

Coq v13 also contains an example

$$z = 2, \quad x = 1, \quad n = 3,$$

where real reconstruction is possible, but integer parameters $m, p \in \mathbb{Z}$ satisfying

$$z = m^3 + p^3, \quad x = m^3 - p^3,$$

do not exist. This is formalized in the lemma

$$\text{no_parameters_for_example}.$$

This aligns with the mathematical idea of the paper: real reconstruction can be formally correct, but if integer parameters are additionally required, further restrictions arise.

10. Consistency of Version Evolution and Final Version v13

Earlier versions of the Coq file verified the core conditional scheme:

if a hypothetical solution implies $o^n = 2n$,

and if it is then accepted that

$$o = 2,$$

then the only possibilities are

$$n = 1 \quad \text{or} \quad n = 2.$$

Later, an explicit real structure was added for the parameters

$$m, p, l, x, y, z$$

and the transition

$$y = ol, \quad y^n = 2nl^n \implies o^n = 2n$$

was formally demonstrated.

Then, the modular caveat was added at the Coq level, formally showing that:

equality in integers

and

modular congruence

are distinct logical objects.

In version v13, three important points were additionally refined.

First, the real factorization of the binomial core is written as

$$\text{core} = n l^n$$

via

real_core_factorization,

which removes the ambiguity between factoring out a real multiplier and integer divisibility.

Second, the local scale $q(j)$ and the global scale $o(n)$ are separated via the structure

LocalScaleGlobalNormalization.

Third, the final form

$$n = 2^{n-1}$$

is also verified via the lemma

binary_scaling_roots_only_one_two.

Thus, v13 does not contradict previous versions; rather, it refines and expands them by adding explicit distinctions:

real parameterization,

real factorization of the core,

integer specialization,

modular caveat,

local scale $q(j)$,

global binary normalization $o(n) = 2$.

11. What Exactly Coq v13 Proves

Coq v13 proves the following propositions.

11.1. Correctness of Elementary Parametric Equalities

If

$$z = m^n + p^n, \quad x = m^n - p^n,$$

then

$$z + x = 2m^n, \quad z - x = 2p^n.$$

In the real case, these are standard equalities over

$$\mathbb{R}.$$

In the integer case, they yield additional parity conditions.

11.2. Correctness of the Real Factorization of the Core

Coq v13 establishes that the notation

$$\text{core} = n \, l^n$$

is understood as a real factorization.

The lemma

$$\text{real_core_factorization_substitutes_y_power}$$

shows that if

$$y^n = 2 \, \text{core},$$

and

$$\text{core} = n \, l^n,$$

then

$$y^n = 2n \, l^n.$$

11.3. Correctness of the Transition from $y = o \, l$ to $o^n = 2n$

If

$$y = o \, l,$$

$$l > 0,$$

and

$$y^n = 2n \, l^n,$$

then

$$o^n = 2n.$$

This is the primary real transition of the model.

11.4. Correctness of the Boundary Normalizing Record

Coq v13 introduces the predicate

$$\text{boundary_q_equation } n \ q,$$

meaning

$$q^n = 2^n.$$

The lemma

$$\text{boundary_q_two}$$

verifies that

$$q = 2$$

indeed satisfies this boundary equation.

The structure

$$\text{LocalScaleGlobalNormalization}$$

then links the non-degenerate region $j > 1$, the boundary value $q = 2$, and the global normalization

$$o = 2.$$

11.5. Correctness of the Final Conditional Exponential Part

If we assume

$$o = 2,$$

then from

$$o^n = 2n$$

follows

$$2^n = 2n.$$

Coq proves that in positive natural numbers, this is only possible for

$$n = 1 \quad \text{or} \quad n = 2.$$

Consequently, for

$$n > 2$$

a contradiction arises.

11.6. Correctness of the Modular Caveat

Coq v13 proves that an integer equality implies a modular congruence, but a modular congruence by itself is not an integer equality.

Formally:

$$x^n + y^n = z^n \quad \implies \quad x^n + y^n \equiv z^n \pmod{q}.$$

But the converse is generally false:

$$x^n + y^n \equiv z^n \pmod{q} \quad \not\Rightarrow \quad x^n + y^n = z^n.$$

This settles the methodological objection claiming that modular examples supposedly refute the real parameterization automatically.

11.7. Minimal p -adic Bracket

Coq v13 also contains the section

PadicBracket.

It introduces the structure

OPadic,

which stores the valuations

$$v_p(o).$$

The definition

padic_equation

writes a condition of the form

$$n v_p(o) = 0$$

for odd parameters $p > 1$.

The lemma

odd_primes_vanish_in_o

shows that given the corresponding condition for $n = 1$ and $n = 2$, the odd p -adic components vanish:

$$v_p(o) = 0.$$

The lemma

two_adic_normalisation

formalizes the normalization

$$v_2(o) = 1$$

from the corresponding condition.

This section should be understood as a minimal formal p -adic block consistent with the idea of binary normalization. It is not a complete theory of p -adic numbers, but it establishes the local algebraic concept that the only preserved normalizing component is associated with the number two.

12. What Coq v13 Does Not Prove

For the sake of intellectual honesty, it is important to explicitly point out what Coq v13 does not prove.

12.1. Coq v13 Does Not Unconditionally Prove $o(n) = 2$

The equality

$$o(n) = 2$$

remains the global binary normalization hypothesis.

Coq proves that if this hypothesis is accepted, the subsequent deduction is correct.

But Coq does not prove that

$$o(n) = 2$$

follows solely from real parameterization.

12.2. Coq v13 Does Not Analytically Prove the Entire Limit $\lim_{j \rightarrow 1+} q(j) = 2$

In the paper, the limit

$$\lim_{j \rightarrow 1+} q(j) = 2$$

is motivated by the binomial identity

$$2 \sum_{\substack{1 \leq k \leq n \\ k \text{ odd}}} \binom{n}{k} = 2^n.$$

Coq v13 fixes the corresponding boundary normalizing record via

boundary_q_equation

and the structure

LocalScaleGlobalNormalization.

But Coq v13 does not construct a full analytical derivation of the limit from the function $q(j)$ as a standalone theorem in analysis.

12.3. Coq v13 Does Not Prove the Full Theory of Finite Fields

The modular section does not build a full-fledged theory of the field

$$\mathbb{F}_q.$$

It does not require q to be prime, does not introduce the multiplicative group of the field, and does not analyze invertible elements.

The modular section serves a more modest and precise purpose: it formalizes the distinction between equality in integers and modular congruence.

12.4. Coq v13 Does Not Prove Unconditional FLT

The file proves a conditional scheme:

if a hypothetical solution for

$$n > 2$$

implies

covers_with 2 n ,

then a contradiction arises.

That is, it proves not the unconditional FLT, but a conditional theorem of the form:

$$(x^n + y^n = z^n \implies 2^n = 2n) \implies \text{there are no solutions for } n > 2.$$

This is exactly the status that should be honestly stated in the paper.

13. General Interpretation of the New Logic

The new logic of paper v13 can be formulated as follows.

First, Fermat's equation is considered

$$x^n + y^n = z^n.$$

Then it is rewritten as

$$y^n = z^n - x^n.$$

Next, the parameterization is introduced

$$z = m^n + p^n, \quad x = m^n - p^n.$$

After this, we obtain

$$y^n = (m^n + p^n)^n - (m^n - p^n)^n.$$

The binomial expansion yields an external binary coefficient

$$2.$$

Then the real scaling core

$$l$$

and the scaling parameter

$$o$$

are introduced.

Given the condition

$$y = o l$$

and

$$y^n = 2n l^n$$

we get

$$o^n = 2n.$$

Next, the local parameter

$$q(j),$$

which arises after the substitution

$$m = j p, \quad j > 1,$$

is considered.

It depends on j and is not identified with the global scale:

$$q(j) \neq o(n).$$

However, on the symmetric boundary side

$$j \rightarrow 1+$$

its normalizing value is

$$\lim_{j \rightarrow 1+} q(j) = 2.$$

This value is adopted as the global binary normalization:

$$o(n) = 2.$$

Then

$$2^n = 2n.$$

This equation has only positive integer solutions

$$n = 1 \quad \text{and} \quad n = 2.$$

Consequently, for

$$n > 2$$

the hypothetical solution leads to a contradiction within the framework of the global binary normalization hypothesis.

The modular caveat simultaneously states that the congruences

$$x^n + y^n \equiv z^n \pmod{q}$$

may have solutions for

$$n > 2,$$

but this is not an objection to the scheme because such congruences are not integer equalities

$$x^n + y^n = z^n.$$

Graphic appendices D and E in paper v13 should be understood as heuristic visualization and geometric motivation for the role of the binary value

$$o = 2.$$

They do not constitute an independent proof of FLT; the rigorous conditional part remains the algebraic reduction to

$$2^n = 2n$$

under the hypothesis

$$o(n) = 2.$$

Final Conclusion. *With the understanding that*

$$m, p, l \in \mathbb{R}$$

the equalities

$$x^n = (m^n - p^n)^n,$$

$$z^n = (m^n + p^n)^n,$$

$$y^n = 2n l^n$$

are correct as a formal real scheme.

Coq v13 consistently formalizes this scheme via real parameters, separately fixing the real factorization of the binomial core, integer specialization, the modular caveat, the local-global distinction between $q(j)$ and $o(n)$, as well as the final exponential-linear equation.

If we then separately accept the global binary normalization hypothesis

$$o(n) = 2,$$

the subsequent deduction

$$2^n = 2n$$

and the restriction

$$n \in \{1, 2\}$$

are correct and machine-verified.

Consequently, under the specified formulation, the core conditional logic is consistent: the weak point lies neither in the real parameterization, nor in the equality

$$y^n = 2n l^n,$$

nor in the modular examples, nor in the graphic appendices, but exclusively in the status of the hypothesis

$$o(n) = 2.$$

Modular congruences for

$$n > 2$$

may exist, but they belong to a different level of the problem and are not integer solutions to Fermat's Last Theorem.

This is exactly what is now explicitly reflected both in the text of paper v13 and in the Coq file v13.